



Perle 3000

Game Design Document

Sofia Strunden | Diploma Project | February 2025

Perle 3000

Run through the Streets with Leo, a Girl, that chases down men that give her unwanted attention. Experience being bold, cheeky and always in charge.

A comedic comment about catcallers, that are always men, and a girl, that is never afraid, that leaves the player with a rewarding feeling of self-empowerment.

Duration: 5min

Format: Mobile Game

Team

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Game Developer: Charlene Masri

Development Support: Justus Henne

Sound Design: Cäcilie Willkommen, Vladyslava Kandyba

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Voice Actors Protagonist: Viola Pröttel, Emma Bading, Sara Keßling

Voice Actors Catcaller: Ilya Barrett, Rocco Häuser, Paul Mandelkow

Production

This mobile game is the Diploma project of Charlene Masri and Sofia Strunden. They both study the course Interactive Media in the specialization as Transmedia Games Artist and Transmedia Games Director at the Animationsinstitut of the Filmakademie Baden-Württemberg in the years since 2022 until April 2025.

Charlene and Sofia produced the game themselves and with the financial support of the Filmakademie Baden-Württemberg.

Project monitoring and coordination

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1 Introduction

1.1 Game Overview

In Perle 3000, we aim to empower players through an engaging, fast-paced experience while tackling the serious topic of street harassment, specifically catcalling. The game's genre, format, and gameplay mechanics have been carefully chosen to create a dynamic, immersive environment where players can actively confront and respond to harassment in an empowering way.

Through the combination of intuitive navigation systems, camera dynamics, and the overarching design choices, we've developed a gameplay experience that reflects the emotional and physical intensity of street harassment and the agency to respond. In this document I will describe the core design elements of the game and explain how each aspect - together with the tools we've employed - aligns with the political topic we are addressing.

1.2 Vision Statement

Our game seeks to empower FLINTA*, by providing an interactive space to confront harassers and reclaim public spaces. For many FLINTA*, when they are read as women, it is often reason enough to be sexualized and confronted with unwanted behavior from men. Harassment is part of their everyday lives.

In our game, we want to highlight the form of harassment that exists in public space in the form of catcalling.¹ Catcalling is usually defined as a rude, derogatory or unwelcome comment, whistle, kissing sound or maybe even a literal meow. But it's more than these passing jeers. This form of street harassment, restricts people's access to public space and can have serious consequences.

¹ Our game focuses on sexist harassment (catcalling) which is one form of discrimination that many individuals, particularly women, face in public spaces. However, we want to emphasize that sexist harassment is just one of many forms of discrimination. While we chose to address this particular issue, we do not intend to suggest that it is more prevalent or more significant than other forms of harassment or discrimination.

Our vision is to offer a game in which players can find an outlet for their anger and frustration about this and instead experience a sense of empowerment by fending off the harassers. Cultivating this feeling will enable the players a sense of relief and makes them feel stronger in their everyday lives. The format of a game allows for the direct involvement of the user, who thus proactively engages in self-determined behavior, leading to a higher self-esteem as a take-away.

Next to the feeling we aim to evoke in the individual user, our goal is also to establish visibility on the topic of catcalling. Deciding to tackle this topic through the format of a game meant staying true to our demands producing media ourselves and accepting the challenge to combine meaningful content with a fun-to-play experience. Our primary target audience is teens and young adults, as this group is not only directly impacted by the dynamics of public space and street harassment but is also highly active in digital spaces and open to engaging with social issues through mobile media. This age group is often in the process of developing their sense of self, social awareness, and empathy, and we believe that this game can serve as both an educational tool and an empowering experience. Games as a medium resonates with young people and offers them the opportunity to experience and reflect on a serious topic they might identify with and learn from.

1.3 Design Philosophy

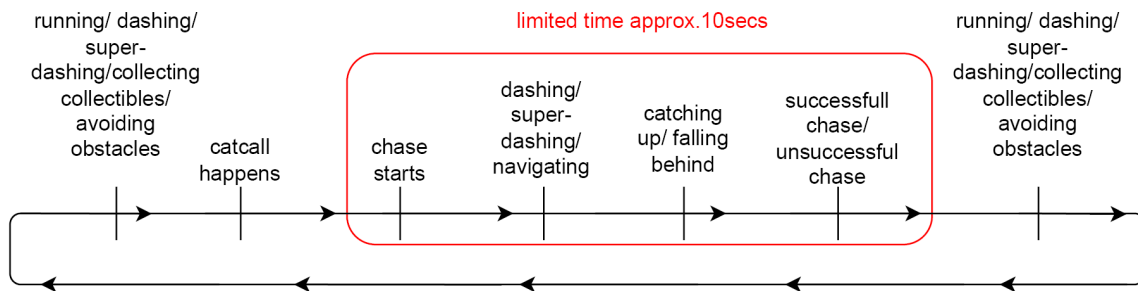
Throughout the development process, we have made an effort to stay informed and up to date with the current debate on harassment. This includes researching current legal definitions, national laws and public opinion on harassment to ensure that our approach remains sensitive and relevant. Our goal is not only to entertain, but also to contribute to a broader dialogue about the impact of street harassment and the various ways to combat it. With these design decisions, we want to create a game that provides an empowering experience while respecting the emotional weight of the topic.

In the context of catcalling it is important to support the range of actions, from ignoring, to chasing back, and value both and everything in-between as a strong response. Due to limited time and team resources, we had to accept that this spectrum of combat

mechanics can only be partially designed and implemented by us and will therefore be features for a future vision concept.

2 Core Mechanics

2.1 Game-Loop



2.2 Mechanics

Perle 3000 Gameplay					
User Interaction	DASH	SUPER-DASH	COLLECTIBLES	OBSTACLES	OPPONENT
	- triggered by pressing A or B - used to accelerate speed	- triggered by holding B - used to accelerate speed - used to destroy	can be collected in environment outside the chase zones	- in environment (balancing: checking if in and/or around the chase zones) - can be avoided through destroying or navigating around	- announces itself through catcall sound - when close enough, chase will start automatically - dash to catch up
Impact	- acceleration - enables successful chase of opponent	> direct acceleration of speed - acceleration of strength - if charged "fully": gives character the strength to destroy obstacles - if used in chase in catch-up moment with opponent: same effect as dash	per collected: reward	- if destroyed: reward - if navigated around: no consequence - if not avoided: slowdown in speed and punishment	- if player doesn't dash, no successful chase - chase successful if: caught up in limited amount of time - if successful chase: reward
Score	successful chase: +3	- successful destroy: +3 - successful chase: +3	reward: +1	- reward: + 3 - punishment: -2	reward: + 3

2.3 Dash mechanics (dash vs. super-dash)

The simple Dash is activated for each time pressing the A- or B-button activating a dash forwards, simulating an acceleration of speed. By button-mashing players can cover more distance in less time and therefore catch up with the opponent in a chase, but also risk crashing into obstacles as they have less reaction time.

The super-dash is activated by pressing and holding the A- or B-button and this provides an indirect acceleration of speed and a significant increase in power. By pressing and holding the button long enough, the super-dash enables the player to break through obstacles, increasing the chances of winning the chase against the attacker, or allowing the player to experience a strong sense of power as they face the challenges in their path.

2.4 Enemy encounter

Enemy encounters are central to the game, serving as a direct reflection of the theme of street harassment. In the environment, opponents take on the role of catcallers, and they announce themselves from a distance through sounds that are easily recognized as verbal harassment - whether through whistles, jeers, or other aggressive exclamations. These sounds act as a signal that an encounter with the catcaller is imminent.

As the player gets closer to the source of the harassment, the catcallers will begin to run away, triggering a chase. The player has the option to either engage in the chase, experiencing a dynamic and rewarding interaction, or pretend chasing them only to scare them off, or not even play along but continue doing their own thing. Chasing down the catcallers rewards the player with an increase in score, reinforcing the action of confronting harassment, while also providing a sense of satisfaction and accomplishment. The chase is built around fast reaction times. To catch up to the fleeing opponent, the player must button-mash the A-button to trigger the dash, quickly closing the distance. The sound design plays a critical role here, with the audio cues accompanying the chase, such as a start sound and faster foot-steps. Visual effects like a vignette and screen shake

when catching up and overjumping the catcaller further enhance the excitement, intensifying the feeling of the chase and making each successful capture feel impactful. This chase mechanic emphasizes the tension of street harassment, giving the player the sense of empowerment in taking control and catching up with the harasser.

2.5 Navigation

The navigation allows the player to use intuitive and easy-to-understand inputs, to adapt the character's path towards or around the challenges that come in their way, always maintaining a fast and agile speed straight to the next challenge.

The movement system in our game is designed to be simple yet precise, offering players enhanced control without the complexity of a joystick or other input devices. This system operates on a linear path with the added freedom of lateral control, allowing players to guide the character side-to-side while maintaining forward momentum.

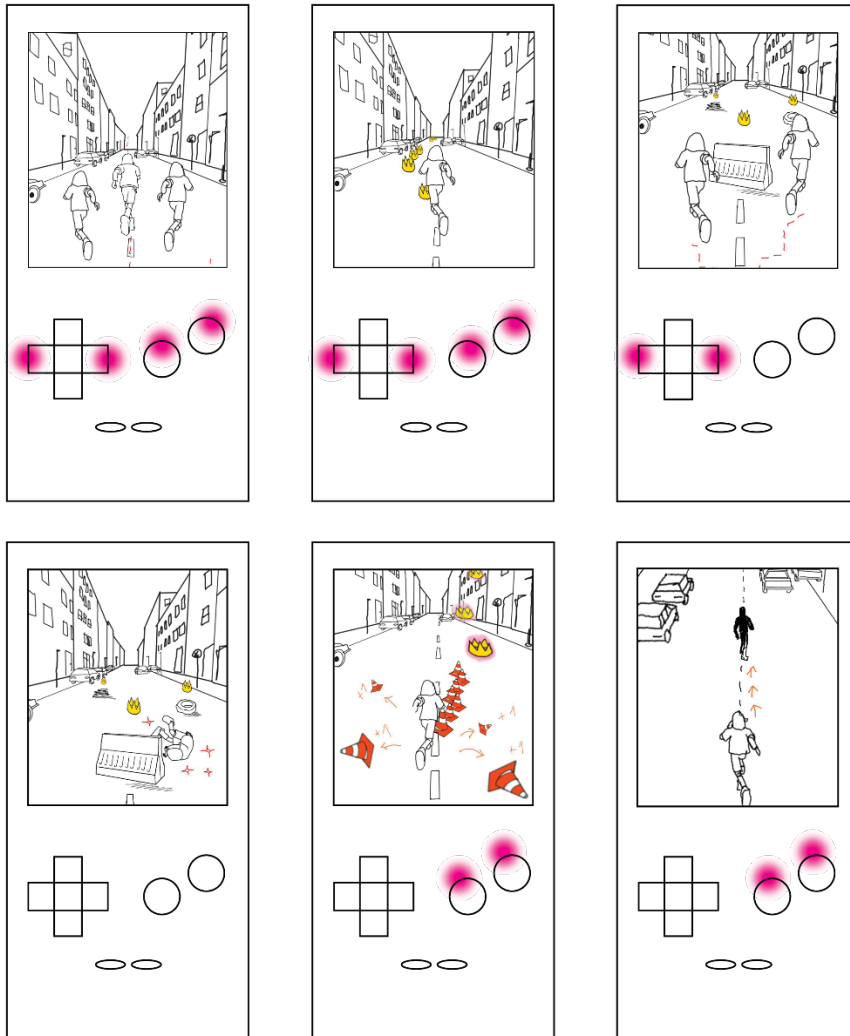
Players can give directional input, controlling the character's lateral movement by tapping or tap and holding the arrow button. When an arrow is tapped, the character rotates in the direction of the arrow (left or right) as long as the button is held, the character will move in that direction. Once the input is released, the character continues to move forward along the linear path, maintaining the horizontal position reached while the input was held. This means that players can dodge or adjust movement while running without losing precision.

This control scheme gives players clear, responsive controls while avoiding the potential frustration of constantly adjusting their direction. The character always progresses forward on the main path, with the player's input acting as a momentary modifier to steer the character left or right, making the system both accessible and engaging.

2.6 Obstacle and destruction mechanic

The obstacles and destruction mechanics allow players to interact with the environment in dynamic ways. As the character races through the streets, they encounter obstacles that

block their path. The super-dash mechanic allows the character to destroy and smash through these obstacles, clearing the way and reinforcing the sense of power and control. The destruction mechanic also serves as a metaphor for breaking down barriers and societal constraints, creating a satisfying feeling of triumph with every obstacle destroyed.



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2.7 Collectible System and Total Score

The core mechanics of the game revolve around empowerment through proving courage. Players are encouraged to interact with their environment, collect items, and confront

² Early state of Perle 3000 core mechanics, effects are not final

obstacles in a way that enhances their sense of control and strength. The game features a global score that rewards the player for their ability to navigate and overcome challenges. While the game rewards players for action and accomplishment, there are consequences for failing to navigate the environment effectively.

Reward, additional points:

- Destroying obstacles: by performing actions like super-dashing through obstacles, overcoming challenges in their path.
- Successful chases: when the player catches up and defeats an opponent, successfully completing the chase sequence.
- Collecting items: by collecting specific items or objectives during the gameplay, encouraging exploration and interaction with the environment.

Punishment, deceleration + point deduction:

- Failing to avoid or destroy an obstacle: by not navigating around an obstacle or not preparing early enough to utilize the super-dash, makes it more difficult to overcome the challenges and catch up with opponents, encouraging players to be strategic.

3 Story and Narrative

3.1 Worldbuilding

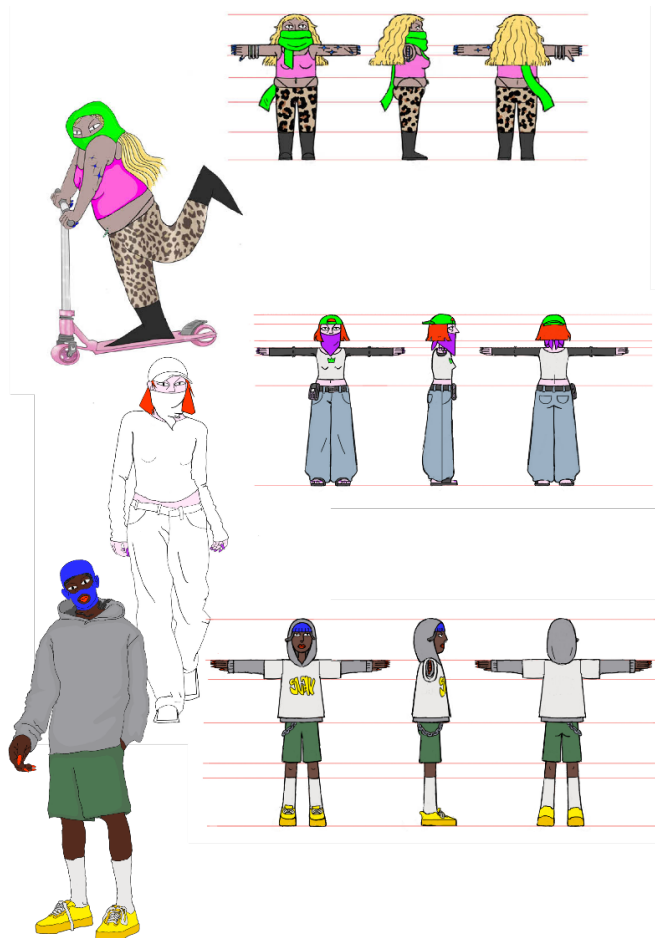
The Game takes place in an urban environment, where the protagonist runs down a rather linear street in a stylized real-world. The game does not cover all various environments, where the player might have already experienced or would expect to experience catcalling, but focuses on this one empty street in daylight which becomes the showplace for the protagonist chasing down catcallers. This setting emphasizes that street harassment doesn't just happen in dodgy places, but mainly in daylight in an environment where women* happen to be for practical reasons, e.g. on the way to the shops or work.



³ In-Game screenshot Perle 3000, not fully set-dressed

3.2 Characters

The Game is to be played with one out of three characters, that are named after big cats: Leo, Jaggy and Puma. There is no particular background story to the characters provided to the Player. Each has an individual style and body statue, bringing signature movements, individual sound-effects and different in-game background music with them. All of the three characters are strong, determined young women* confronting and chasing down rude, inappropriate men who represent the catcalling and harassment that many girls* and women* face in real life. We aim to represent young women* as heroines who stands up for themselves and have the courage to actively fight back, with the goal that many girls* as users, sympathize with them.



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⁴ Concept Art and character exploration

3.3 Meaningful female representation

When we looked at GTA, as one of our main references, in particular, we found a glaring issue: the lack of meaningful female representation. In most cases, women were either absent or highly sexualized and served as non-relevant side characters. As in film culture, the male gaze was (and still is) a trend in games and is responsible for viewing habits and social norms in sexist interactions.

This stark contrast to the needs of many young women* wanting to identify with a strong protagonist and experience a story that they have experienced, inspired us to create a game where women are the main character, courageous, and capable of taking control. By making the protagonist a girl* and her opponents men who act in ways that deserve consequences, we aim to deliver a shift in perspective and narrative. This is a story where young women* can see themselves as powerful figures who take action, and where the message is clear: harassment is not acceptable, and they have the right to stand up for themselves. The game provides not just a fictional escape but a form of empowerment and reclamation of space in a world that often marginalizes women's voices.



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⁵ WIP Screenshot to capture an image to convey Leos emotion

3.4 90's narrative

The 90's aesthetic plays a key role in this narrative. First, we chose it because we believe this kind of game should have existed already back then - providing young women with a fun, action-packed game that lets them be the hero. Second, this style is enjoying a resurgence in popularity today. People are drawn to the style for its simplicity, charm, and nostalgia, which makes it the perfect visual language for our project. We aim to strike a balance between high readability and a signature look, using bright, distinct colors and stylized environments that feel true to the time period while offering a fresh perspective.



4 Art and Visual Design

4.1 Retro Influence

The art style in Perle 3000 is a blend of Game Boy-era graphics, 90's PS1 aesthetics, and early 3D motion graphics. This intentional that evokes a time when personal gaming was just beginning to be accessible to children, particularly in the 90s, when devices like Game Boy and the first consoles made gaming personal. It references an era where gaming became more than just a pastime for a specific group - it became part of everyday life for many, whose families could afford it and were open to it.

In designing the gameplay mechanics, we drew inspiration from fast-paced, low-poly action racing games like Road Rash, Pepsi Man, and Grand Theft Auto (GTA). These games were not just fun to play, but their visuals and mechanics resonated with us while we considered the premise of a girl fighting back against catcallers. Games like these had a straightforward, accessible approach to action gameplay, and their focus on speed and challenge made them ideal touchstones.



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⁶ Screenshot from Grand Theft Auto 3 <https://www.gamestar.de/artikel/gta-3-im-test-die-3d-revolution-der-grand-theft-auto-serie,1338146.html>

4.2 Character and Environment Design

Characters and environments are designed with minimal polygons to ensure clarity and performance, while staying visually interesting. To stay true to the retro aesthetic, we project textures and drawings onto low-poly UV maps, like the buildings that are on the side of the road, or the clothes. A dithering shader is applied to both characters and environments to simulate gradients and textures in a way that fits the retro aesthetic, reminiscent of older hardware limitations.

To complement the 3D models and add to the retro feel, we use 2D effects such as pixelated particle systems (e.g., sparks, dust, and impact effects). These effects retain the game's nostalgic aesthetic and help accentuate player actions (e.g., successful chases, super-dash).

The retro color palette and stylized shapes emphasize the game's nostalgic feel while maintaining modern 3D elements for smooth performance on mobile devices.



⁷ UV Map of Leo

⁸ Character Leo in the process of being Textured



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⁹ Concept Art for the house blocks in the environment

¹⁰ Reference: Road Rash 1 <https://www.videospielgeschichten.de/road-rash-knuppel-dich-ins-ziel/>

4.3 UX/ UI Design

Combining low-poly 3D models and a dithering shader, inspired by the retro aesthetics of early portable games that were played on consoles like for example the Game Boy 1 and Game Boy Color, led us to the decision to base our Interface Design and Input system in reference to this Design era.

The HUD and menu feature stylised UI elements, 2D icons and typography that stays true to the Game Boy aesthetic. The characteristic photorealistic overlay of a Game Boy Color frames the game screen and gives the feeling of playing on a real handheld device.

Simple, flat UI elements help maintain clarity and avoid visual clutter, allowing the player to focus on the game. The menus and HUD are kept minimalist yet functional, with pixelated icons and clear typography that reflect the game's retro roots.



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¹¹ Perle 3000 Purple edition, Game menu HUD

¹² Perle 3000 Transparent edition, In-Game HUD

5 Audio and Sound Design

5.1 Sound Effects

For the audio design of our game, we've found it very important to design it at an early point to achieve to create a dynamic and engaging soundtrack and sound effects that complement the game's atmosphere while not being too explicit and thus trigger negative emotions, but still are remarkable enough to be recognised as catcalls in a public space environment. For the sound design, we partnered with two sound artists to create a mix of artificial, foley, and voice sound effects. The sound effects include interface sounds and dynamic effects, like footsteps, that guide the gameplay, ambient sounds, and wallas to create the feeling of a public environment and help immerse the player in the world. For the sound effect featuring voice actors to simulate the catcall we have conceptualized a wide variety in the aspects of explicitness, offering a range of tones and intensities to and decide through testing with test-players to ensure it feels realistic without crossing into discomfort.

These vocalizations vary in:

- Explicitness: From non-explicit mumbling to more explicit language
- Distance: From shouting to more subtle self-talk
- Aggression: From rude remarks to responses such as fear and crying

5.2 Music

For the music, we worked with a musician to create three distinctive soundtracks in different genres: Jungle, Hip-Hop and Techno. Each genre offers an individual mood to match the different characters and personalities of the users of the game. In addition to the instrumental versions, two of these tracks also feature vocals, written and sung by another artist we collaborated with.

6 Meta Progression

6.1 Level Design

Each level in the game is structured with a countdown timer signaling the start. The gameplay then progresses over a few minutes, during which the player navigates/ super-dashes obstacles, chases opponents, and collects items. The level ends with a finish line, where the player's total score is calculated. Afterward, players can view their stats and compare scores to previous rounds, with the goal being to achieve a high score to unlock the next level.

6.2 Player Progression

Player progression is primarily driven by achieving high scores across levels. The player's score is tracked and accumulated, with points awarded for:

- Super-dashing through obstacles
- Dashing to catch up with opponents
- Defeating opponents
- Collecting collectibles

The highest score for each level is saved, allowing players to strive for better results in subsequent rounds. By continually improving their performance, players unlock new levels, motivating them to keep progressing through the game

7 Technical Aspects

7.1 Format

The action/arcade runner genre is ideal for providing a tense experience centered around speed, quick reflexes and constant movement. By having the character chase after harassers in a fast, linear environment, we ensure that the gameplay reflects the feeling of being targeted or pursued – but switch the perspective. The third-person camera allows players to be fully immersed in the character's world, where they control the protagonist's movements and actions.

To make the gameplay feel even more immersive and rewarding, we've included screen shakes that are triggered when the player encounters an obstacle or defeats an enemy. The screen shake amplifies the physical impact of the character's actions, whether it's the force of an impact or the high of defeating an enemy. This sensory feedback not only makes the game more dynamic, but also increases the satisfaction of overcoming obstacles or defeating catcallers.

7.2 Platform

We chose to make Perle 3000 a mobile game for accessibility reasons. By bringing this game to mobile platforms, we ensure that it is more accessible since mobile devices are portable, always within reach, and more affordable, making the game accessible to a broader range of players. We want this message of empowerment to be available to as many people as possible, and mobile gaming provides an ideal platform to reach and engage more players.

8 Conclusion

8.1 Iterative Design

Throughout development, we went through many iterations and versions, refining ideas and mechanics. As a team with limited experience in game design, we relied heavily on frequent user tests with fellow students and friends. These quick feedback loops allowed us to quickly identify what worked and what didn't, leading to key improvements; such as shifting from a joystick-based input system to the current lateral control input system, which significantly improved player control and engagement.

8.2 Final Product Reflection

I am proud of what we have achieved and I think that it is a solid starting point for making games. While it was a challenging process, I learned a lot regarding team work, communication, honest self-estimation and commitment. Working with my collaborator Charlene on this project was, by far, the best part of this project. The overarching theme naturally made our workspace a safe space for sharing experiences and it was special to have this place every day, where I knew we had an outlet for the frustration and anger. I myself experienced a boost in confidence in interacting with catcallers; and hope that the user of this game experience the same.

8.3 Future Vision

Looking forward, our goal is to balance the game's core mechanics and expand features like environments, level and character that we polish to ensure an engaging, smooth experience. Once we achieve that, we aim to publish the game to reach as many players as possible, making it accessible to a broad audience.

The best-case scenario is to see the game enjoyed by many and, ideally, generate revenue to support further development and expansion.